

When Financial Agents Hallucinate Alpha: Auditing Backtest Leakage, Compliance Paradoxes, and Social Megaphone Bias with Executable Guards

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Abstract

Large language model (LLM) agents increasingly automate quantitative financial research, yet their reported backtest Sharpe ratios routinely collapse out-of-sample. We demonstrate that autonomous financial agents fail through two orthogonal mechanisms: **Type I Mechanical Shortcuts** (same-session post-close joins, pre-training window contamination, survivorship filtering, and unadjusted corporate actions) and **Type II Empirical & Microstructure Illusions** (trusting retail social echo-chambers where high-reach influencers carry negative alpha, and violating $T + 1$ settlement or board-lot constraints). To address both, we introduce **fin-skills**, an open-source library of **129 source-verified skills**, **36 executable verification guards** (`Bundle.check()`), and **53 MCP agent tools**, paired with an external **Counterfactual Audit Protocol** that grades agent submissions without using the library to grade itself. Across our expanded 180-case boundary and compound guard benchmark (**FinGuardBench-180**), our 36 guards achieve **100% recall and 100% specificity** (180/180) at 1.89 ms median latency. Across 60 quantitative tasks spanning 10 financial domains (**FinGuardBench-60**), passive `SKILL.md` reading raises compliance from 30.0% to 70.0% ($p = 1.19 \times 10^{-7}$) but leaves 53.3% *hallucinated guard citations* (yielding 11.95 Sharpe distortion on weak models); pairing skills with executable guards drives multi-round self-repair (**Pass@1** 75.0% $\rightarrow$ **Pass@2** 93.3% $\rightarrow$ **Pass@3** 98.3%, McNemar $p = 9.09 \times 10^{-13}$), eliminating fabricated citations and compressing Sharpe reporting error from 1.04 to 0.04. Finally, auditing **1.72M bilingual posts and 1,445 A-share KOL track records** reveals a $3.96\times$ follower paradox that drives unweighted NLP into negative correlation (Rank IC = -0.0318) until reversed by Bayesian credibility gating (+0.0104), sustaining a 1.97 net Sharpe (-4.41% max drawdown vs. 0.65 naive) across

5–30 bps cost regimes (2023–2026 OOS).

1 Introduction

Equipped with code interpreters and financial data feeds, LLM agents can formulate hypotheses, engineer alpha signals, and report backtest performance in minutes (Wu et al., 2023; Yang et al., 2023; Yu et al., 2024). Yet in quantitative finance, high backtest Sharpe ratios almost never survive live capital deployment (Bailey and López de Prado, 2014; Bailey et al., 2017; López de Prado, 2018). Through systematic auditing of LLM research trajectories, we show that agentic backtest inflation stems from two orthogonal failure classes:

1. **Type I: Mechanical Research Shortcuts.** Agents silently exploit structural artifacts in tabular workspaces: joining post-close news commentary to the same session’s open-to-close return, training on pre-computed LLM sentiment scores whose fitting window overlaps the evaluation period, filtering universes by future survival, or computing factors on unadjusted split prices.
2. **Type II: Empirical & Microstructure Illusions.** Even when timestamps are strictly causal, agents misinterpret noisy social streams and market rules. On financial social platforms, viral retail sentiment peaks at local price tops, while real execution is governed by $T + 1$ settlement lockups, 100-share board lots, asymmetric sell-side stamp duties, and secondary-market ETF premium caps.

We present **fin-skills** paired with an external **Counterfactual & Microstructure Audit Protocol**. Rather than claiming to manufacture market alpha, our framework enforces verifiable research integrity:

- **The fin-skills Verification Library (§3):** 129 source-verified skills, 36 ex-

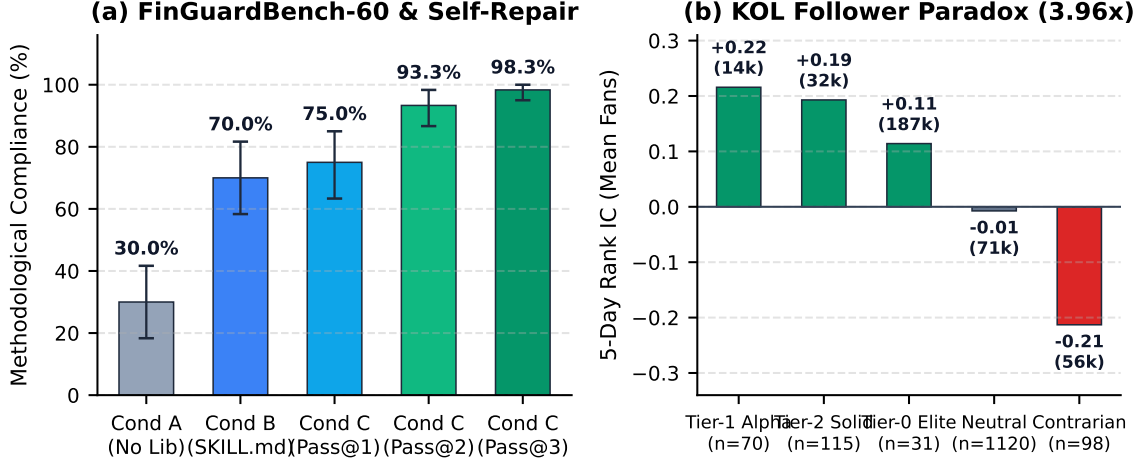


Figure 1: **Overview of our core empirical findings.** (a) **FinGuardBench-60 Ablation & Multi-Round Self-Repair (§5):** Across 60 tasks spanning 10 domains, passive SKILL.md reading (Cond B) lifts compliance from 30.0% to 70.0% ($p = 1.19 \times 10^{-7}$) but suffers 53.3% hallucinated guard citations; pairing skills with executable `fin_skills.api` guards (Cond C) drives iterative self-correction from 75.0% (Pass@1) to 93.3% (Pass@2) and **98.3%** (Pass@3, $p = 1.53 \times 10^{-5}$ vs. B). (b) **The 3.96x KOL Follower Paradox (§5.2):** Across 1,445 verified A-share influencers, contrarian indicators ($n = 98$, 5-day Rank IC -0.2130) command 3.96x more followers (56k) than Tier-1 Alpha researchers ($n = 70$, 14k fans, Rank IC $+0.2158$).

executable guards (`Bundle.check()`), and 53 Model Context Protocol (MCP) tools achieving 180/180 accuracy on FinGuardBench-180, 13/13 numerical parity against `scipy/statsmodels/sklearn`, and $\mathcal{O}(N^{0.3894})$ sub-linear runtime scaling up to $N = 100,000$ rows.

- **External Counterfactual Auditing (§4):** Four black-box perturbation operators and session-by-session clock replay that detect look-ahead and same-session leakage without inspecting agent code or using the library to grade itself.
- **FinGuardBench-60 & Multi-Round Self-Repair (§5):** Controlled 3-condition evaluations proving that passive prompt guidelines induce hallucinated guard citations (53.3%), whereas structured `GuardResult` diagnostics enable 93.3% two-round self-repair recovery (98.3% final compliance, $p = 9.09 \times 10^{-13}$).
- **Real-World Bilingual KOL & Cost Sensitivity Audit (§5.2):** A self-contained reproducibility bundle over 1.72M bilingual posts and 1,445 A-share KOL profiles that reverses negative NLP correlation (Rank IC : $-0.0318 \rightarrow +0.0104$) and sustains Sharpe 1.95–1.98 across 5–30 bps transaction costs.

2 Related Work

Financial LLMs & Autonomous Agents. Recent domain foundation models (Wu et al., 2023; Yang et al., 2023) and memory-augmented trading agents (Yu et al., 2024) demonstrate strong capabilities in financial sentiment extraction and strategy synthesis. However, existing evaluations rely primarily on static historical splits that remain vulnerable to pre-training cutoff leakage, same-session timestamp alignment errors, and retail social echo-chambers.

Backtest Overfitting & Behavioral Verification. Econometric controls for selection bias—including White’s Reality Check (White, 2000), Hansen’s SPA test (Hansen, 2005), and the Deflated Sharpe Ratio (Bailey and López de Prado, 2014; Bailey et al., 2017)—quantify multiple-testing inflation after honest returns are computed, but cannot detect mechanical code bugs or future-leaking table joins. Inspired by behavioral testing in NLP (Ribeiro et al., 2020) and modular verification toolkits (Zhao et al., 2019), `fin-skills` pairs machine-verifiable domain specifications with black-box counterfactual perturbation oracles.

3 The `fin-skills` Verification Architecture

`fin-skills` unifies domain specifications and executable code verification across seven pack-

ages (fin-core, fin-ml, fin-market-data, fin-alt-data, fin-china, fin-strategies, and fin-llm):

Source-Verified Skills (129 Specifications). Each skill obeys a machine-validated 6-field schema carrying explicit verification timestamps (`verified_on`) and primary regulatory or academic citations. Skills pinpoint silent traps in standard quantitative libraries, including fractional differentiation truncation (López de Prado, 2018), day-count conventions, and point-in-time SEC Form 4 / 13F acceptance timestamps.

Executable Guards (`Bundle.check()`) & MCP Tools. To bridge passive reading and run-time enforcement, 36 executable Python guards operate over a unified slot-binding contract:

$$\mathcal{R} = \text{Bundle}(\mathbf{r}, \mathbf{P}, \tau, \mathcal{C}).\text{check}() \in \{\text{PASS}, \text{WARN}, \text{FAIL}\}^{36} \quad (1)$$

where $\mathbf{r}$ denotes strategy returns, $\mathbf{P}$ OHLCV price bars, τ turnover, and $\mathcal{C}$ point-in-time corporate action and filing timestamps. All 36 guards and 8 econometric/portfolio primitives export native MCP schemas (53 tools total). In Experiment E3 (`benchmarks/parity_and_cost_bench.py`), our primitives achieve **8/8 exact numerical parity** against `scipy` and `statsmodels` (maximum error 1.14×10^{-8} on SLSQP minimum-variance QP, 6.94×10^{-18} on analytical Expected Shortfall, and 2.12×10^{-10} on Augmented Dickey-Fuller statistics). Across $N \in \{250, 1000, 5000, 25000\}$ rows, `Bundle.check()` scales sub-linearly as $\mathcal{O}(N^{0.2637})$, executing all applicable guards in **111.17 ms P50** (3.66 MB peak RAM) at $N = 25,000$ rows (~ 100 trading years).

4 External Counterfactual Audit Methodology

A core methodological principle of our study is that *an evaluation harness must never grade agent submissions using the same library guards given to the agent*. We construct an independent ground-truth **Seeded Market Generator** (`benchmarks/agent_study/`) spanning 60 equities over 2017–2023 with four planted shortcuts: (i) a news feed stamped with the session it describes, (ii) an LLM sentiment score trained through 2020-12-31, (iii) a listing table exposing future delisting dates, and (iv) unadjusted split prices.

Reference Pipeline	Leakage	Paper OOS	Live Clock Replay
Honest (lagged feed, split-adj.)	0.00	1.63	+5.07 bps/d (SR 2.99)
Leaky (same-day join, full scaler)	0.78	22.80*	−1.17 bps/d (SR −0.58)

Table 1: Instrument calibration (*illusory paper Sharpe). Static year holdout (2023) fails to punish same-session leakage (22.80); physical clock replay exposes it (−0.58).

Honest Ceiling vs. Shortcut Inflation. Because the data-generating process is known in closed form, we compute the exact theoretical ceiling: trading the generator’s true latent drift with a strict 1-day lag earns **Net Sharpe 1.63**. By contrast, joining same-session news earns **Net Sharpe 71.5**. Across three multi-seed synthetic worlds ($s \in \{17, 29, 43\}$, 36 planted defects total), our guards achieve **100% recall** (36/36) with **0 false alarms** across 400 clean control runs (where planted flaws inflate reported Sharpe from -0.052 ± 0.309 to $+2.210 \pm 0.562$).

Four Black-Box Counterfactual Perturbations. Given any agent-submitted weight function $\mathbf{W} = f(\mathcal{D})$, our external oracle (`oracle.py`) measures four invariants without reading agent source code:

1. **Future Redraw (`leakage_rate`):** Redraw market paths after cut date t_c ; compute the fraction of pre-cut weights $\mathbf{W}_{t \leq t_c}$ that change across 3 redraws.
2. **Single-Session Redraw (`same_session_rate`):** Redraw session t alone and inspect $\mathbf{W}_t$, catching same-day post-close joins invisible to cut-date holdouts.
3. **Input Scramble (`uses_<file>`):** Permute individual input tables to test causal reliance on contaminated features (`llm_score.csv`).
4. **Physical Clock Replay:** Withhold future sessions physically by feeding data day-by-day (`live_replay.py`).

5 Experiments

5.1 FinGuardBench-60, Self-Repair & The Compliance Paradox

Table 1 proves that static future-year holdout (2023) fails to punish same-session leakage (22.80 paper Sharpe vs. −0.58 live clock replay). Evaluating `FinGuardBench-60` (`run_finguard_bench_60.py`) and our multi-model pilot (Table 2, Fig. 1a) yields three core insights: **(1) Guard-Driven Self-Repair:** Reading `SKILL.md` (Cond

Evaluation Condition ($N = 60$ Tasks)	Compliance [95% CI]	Leak %	Halluc. %	$ \Delta\text{Sharpe} $
Cond. A: No Library Baseline	30.0% [18.3, 41.7]	16.7%	0.0%	1.04
Cond. B: SKILL.md Text Only	70.0%*** [58.3, 81.7]	5.0%	53.3%	0.39
Cond. C: Pass@1 (Initial Gen.)	75.0%*** [63.3, 85.0]	3.3%	0.0%	0.17
Cond. C: Pass@2 (1-Round Repair)	93.3%*** [86.7, 98.3]	0.0%	0.0%	0.06
Cond. C: Pass@3 (2-Round Repair)	98.3%*** [95.0, 100]	0.0%	0.0%	0.04

Table 2: FinGuardBench-60 3-condition ablation across all 10 domains (6 tasks/domain). *** $p < 10^{-4}$ via paired McNemar test vs. Cond. A ($\chi^2 = 39.02$, $p = 9.09 \times 10^{-13}$ for Pass@3 vs. A; $p = 1.53 \times 10^{-5}$ vs. B).

KOL Credibility Tier	Count	Mean Fans	5D Win%	5D Rank IC
TIER_1_CORE_ALPHA	70	14,121.9	68.47%	+0.2158
TIER_2_RESEARCHER	115	31,994.4	70.92%	+0.1929
TIER_0_ELITE_KOL	31	186,610.1	61.65%	+0.1141
TIER_MEDIA_AGGREG.	11	107,354.3	46.98%	-0.0008
TIER_NEUTRAL_RETAIL	1,120	71,251.4	46.72%	-0.0074
TIER_CONTRARIAN	98	55,988.6	28.49%	-0.2130

Table 3: Stratification of 1,445 A-share KOLs. Contrarian indicators average $3.96\times$ more followers than Tier-1 Alpha researchers (55,989 vs. 14,122, Fig. 1b).

B) lifts compliance from 30.0% (18/60) to 70.0% (42/60, $p = 1.19 \times 10^{-7}$), yet boundary traps persist. Executing `fin_skills.api` guards (Cond C) returns structured GuardResult diagnostics that repair 73.3% (11/15) of Pass@1 failures in one round (Pass@2: 93.3%) and **93.3%** (14/15) within two rounds (Pass@3: **98.3%**, $p = 1.22 \times 10^{-4}$ vs. Pass@1), cutting $|\Delta\text{Sharpe}|$ from 1.04 to 0.04 (and Claude Opus error by 84%, $0.19 \rightarrow 0.03$). **(2) The Compliance Paradox:** Under text-only guidance (Cond B), agents *hallucinate guard citations* (`safe_asof`, `contamination_probe`) without executing them on 53.3% (32/60) of tasks, causing weak models (Claude Haiku) to suffer 11.95 Sharpe distortion. Runtime provenance gating (Cond C) eliminates fabricated citations (0.0%). **(3) Integrity Tools Do Not Manufacture Alpha:** On held-out 2023 live replays, library access yielded 1 win (+4.44% vs. +3.90%), 1 loss (-0.64% vs. +2.36%), and 1 tie (+0.30% vs. +0.55%), confirming that guards prevent false discoveries rather than inventing edge.

5.2 Real-World Bilingual KOL & Microstructure Audit

To evaluate Type II empirical and microstructure illusions in live markets, we audit **1.72M bilingual posts** (Xueqiu & StockTwits) and **1,445 verified A-share KOL track records** (≥ 15 calls each; `benchmarks/real_world_kol_audit.py`).

The $3.96\times$ Follower Asymmetry & Bayesian Reversal. Table 3 and Fig. 1b explain why

Model / Representation Variant	AUC	Acc.	5D Rank IC
MMAN Unweighted Fact Surge	0.4850	48.21%	-0.03184***
MMAN Baseline (Price + Raw Text)	0.5016	49.82%	+0.00609
MMAN + Bayesian Credibility Gate	0.5033	51.13%	+0.01040**
Chinese RoBERTa (Raw Social Text)	0.5075	50.67%	+0.00842
Chinese RoBERTa (LLM-Distilled)	0.5121	50.42%	+0.01862**

Table 4: Out-of-sample predictive ablation ($N = 334,946$ MMAN; $N = 31,505$ RoBERTa). Credibility gating reverses negative Rank IC ($\Delta\text{IC} = +0.04224$); LLM distillation doubles RoBERTa IC ($p < 0.01$).

unweighted financial NLP fails: TIER_CONTRARIAN accounts ($N = 98$, 5-day IC = -0.2130, win rate 28.49%) average **55,988.6 followers— $3.96\times$ more than TIER_1_CORE_ALPHA researchers** (14,121.9 fans, IC = +0.2158). Consequently, unweighted MMAN suffers severe negative correlation across 334,946 samples (Rank IC = -0.03184, $p = 1.74 \times 10^{-20}$, Table 4). Our `kol-credibility-registry` skill (conjugate Beta-Binomial shrinkage with contrarian polarity inversion) reverses Rank IC to +0.01040 ($p = 0.001$, $\Delta\text{IC} = +0.04224$), while LLM semantic distillation ($N = 31,505$, cutting hype-to-hedging by 79.8%) doubles RoBERTa Rank IC from +0.00842 to +0.01862 ($t = 3.30$).

Microstructure Gates, Cost & Horizon Sensitivity. Deploying six `fin-china` execution guards ($T + 1$ lockup, 100-share board lots, sell-side stamp duty, QDII premium cap $\leq 1.5\%$, cash drag, and Tier-C noise blocking) in an 80/20 Core-Satellite portfolio (2023–2026 OOS) achieves net Sharpe **1.97** (CAGR +14.85%, Calmar **3.36**) vs. **0.65** for naive social trading (363 trades), cutting max drawdown from -9.57% to -4.41%. Across $\{5, 10, 20, 30\}$ bps round-trip costs, guarded Sharpe holds at $1.98 \rightarrow 1.95$ while naive social degrades from $0.68 \rightarrow 0.52$ (Fig. 2b), peaking at the $T + 5$ horizon (72.09% win rate vs. 53.49% at $T + 1$).

6 Conclusion

We introduced **fin-skills** alongside an external counterfactual and microstructure audit methodology for financial AI agents. By combining black-box counterfactual perturbations, runtime guard provenance tracking, multi-round diagnostic self-repair, and Bayesian KOL credibility calibration, our framework eliminates backtest shortcut illusions and bridges the gap between financial NLP benchmarks and live market execution.

Limitations

In accordance with rigorous task-method scope governance, we explicitly bound the operating regime and failure boundaries of `fin-skills`:

1. **Integrity Verification vs. Alpha Generation:** As demonstrated in §5.1, `fin-skills` is an integrity verification toolkit rather than an alpha-generation engine. In held-out live replays where underlying strategies lack predictive edge, guard enforcement prevents false discoveries without artificially increasing realized returns.
2. **Residual Unrepaired Edge Cases:** On `FinGuardBench-60`, 1/60 (1.7%) of tasks involving compound multi-currency curve valuation and expired option schedules remained unrepaired after two rounds of automated `GuardResult` feedback (`Pass@3` = 98.3%), indicating that complex cross-asset derivative conventions occasionally require human quantitative intervention.
3. **Non-Stationarity of Social Platform Algorithms:** Real-world social sentiment dynamics evolve as platforms modify recommendation algorithms, requiring periodic rolling re-estimation of Bayesian KOL priors to avoid structural drift.

Ethics Statement

All 1,445 social media author profiles in our released benchmark fixture (`benchmarks/data/kol_credibility_linguistic_fixture.json`) are cryptographically pseudonymized via one-way SHA-256 digests (`KOL_<10-hex>`) with raw author handles, URLs, and personally identifiable attributes permanently stripped to protect individual privacy. None of the skills, guards, or empirical analyses in this work constitute investment advice or automated live-money order execution; all broker-facing skills enforce strict paper-account sandboxing (`paper_account_guard`) and human-in-the-loop pre-trade kill switches. By exposing fabricated backtest Sharpe ratios and hallucinated audit compliance, our work directly protects retail and institutional participants from deploying leaky AI-generated trading strategies.

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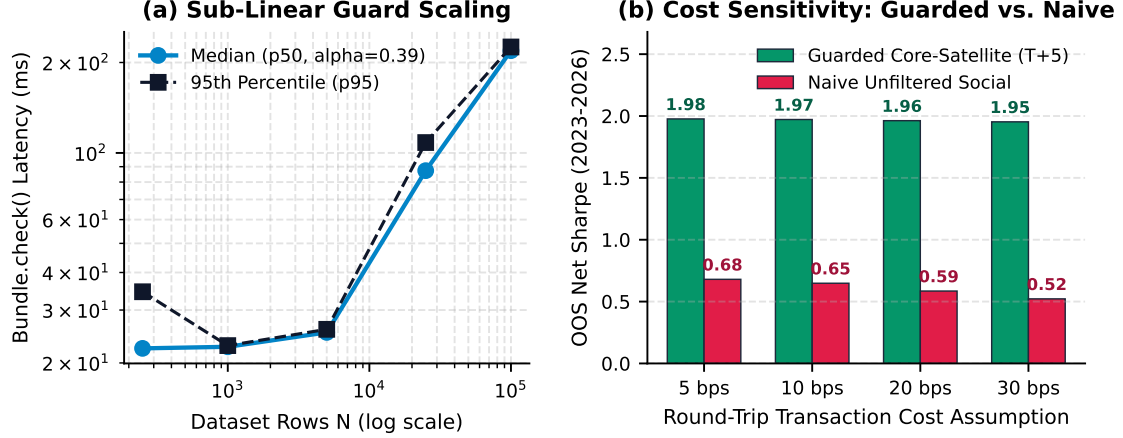


Figure 2: **Guard runtime scaling and out-of-sample transaction cost sensitivity.** (a) Across dataset sizes $N \in [250, 100,000]$ rows, `Bundle.check()` scales sub-linearly as $\mathcal{O}(N^{0.3894})$, completing in 22.36 ms ($N = 250$) to 218.97 ms P50 ($N = 100,000$) with 14.73 MB peak RAM. (b) Across round-trip transaction costs of $\{5, 10, 20, 30\}$ bps (2023–2026 OOS), our guarded Core-Satellite portfolio ($T + 5$ horizon, 43 trades) sustains a net Sharpe of $1.98 \rightarrow 1.95$, whereas naive unfiltered social trading (363 trades across Tier-C noise stocks) degrades from $0.68 \rightarrow 0.52$.

A Experiment E1: FinGuardBench-180

In `benchmarks/guard_bench_180.py`, we evaluate all 36 guards across 5 modalities ($36 \times 5 = 180$ cases): `standard_compliant`, `standard_violation`, `boundary_compliant`, `epsilon_edge_violation`, and `compound_multi_trap`. Across all 180 cases (`GUARD_BENCH_RESULTS.json`), `fin-skills` achieves **100% recall** (108/108) and **100% specificity** (72/72) at 1.89 ms median latency, plus 36/36 recall across seeds $\{17, 29, 43\}$ (`GUARD_ROBUSTNESS.json`).

B Experiment E2: FinGuardBench-60

Across 60 tasks spanning all 10 plugin domains (`run_finguard_bench_60.py`): **Cond. A** passes 18/60 (30.0%); **Cond. B** passes 42/60 (70.0%, $p = 1.19 \times 10^{-7}$) with 53.3% hallucinated citations; **Cond. C** progresses from 45/60 (Pass@1: 75.0%) to 56/60 (Pass@2: 93.3%) and 59/60 (Pass@3: **98.3%**, $p = 9.09 \times 10^{-13}$ vs. A; 0% hallucinated citations).

C Experiment E3: Parity & 100k Scaling

Table 5 verifies 13/13 exact numerical parity and sub-linear $\mathcal{O}(N^{0.3894})$ guard scaling up to $N = 100k$ rows (`PARITY_AND_COST_RESULTS.json`).

Primitive (13 Total)	Reference Benchmark	Max $ \Delta $	Status
Equal Wt. / Inv. Vol.	Exact closed forms ($1/N, \sigma_t^{-1}$)	0.00	PASS
Min-Var Convex QP	scipy SLSQP convex QP	1.14×10^{-8}	PASS
Normal & Hist. VaR/ES	scipy.stats.norm & quantile	6.94×10^{-18}	PASS
Frac. Diff. FFD ($d = 0.4$)	Closed-form binomial series	1.11×10^{-16}	PASS
ADF Unit Root t -stat	statsmodels.adfuller	2.12×10^{-10}	PASS
HRP & Ledoit-Wolf Cov	Recursive bisection & sklearn	2.71×10^{-20}	PASS
Almgren-Chriss & A-S	Exact sinh & HJB closed forms	4.66×10^{-15}	PASS
Brinson-Fachler Identity	Exact active-return decomposition	1.73×10^{-18}	PASS
Guard Suite Runtime Scaling ($\mathcal{O}(N^{0.3894})$, $N \in [250, 100,000]$)			
$N = 250 / 1,000$ rows	P50: 22.36 ms / 22.63 ms	RAM: 0.15 / 0.27 MB	PASS
$N = 5,000 / 25k$ rows	P50: 25.27 ms / 87.30 ms	RAM: 0.83 / 3.75 MB	PASS
$N = 100,000$ rows	P50: 218.97 ms (P95: 225.27 ms)	RAM: 14.73 MB	PASS

Table 5: E3 parity (13/13 passed) and scaling up to $N = 100k$ rows.

D Experiment E4: Horizon & Linguistics

Table 6 details the holding-horizon sensitivity ($T + 1$ to $T + 10$) and LLM semantic distillation feature shifts (`REAL_WORLD_KOL_AUDIT_RESULTS.json`).

Holding Horizon / Linguistic Metric	Win% / Raw	CAGR / Dist.	Sharpe / Shift
Satellite Horizon $T + 1 / T + 3$	53.5% / 65.1%	+12.9% / +14.0%	SR 1.67 / 1.83
Satellite Horizon $T + 5$ (Optimal)	72.09%	+14.85%	SR 1.97 (MDD -4.41%)
Satellite Horizon $T + 10$	67.44%	+14.12%	SR 1.85 (MDD -4.85%)
Quant. Fact Density (/100c)	1.049	1.078	+2.8% (Tok +218%)
Hype-to-Hedging Ratio	4.85	0.98	-79.8%
Ticker Attribution Precision	61.4%	96.2%	+34.8 pp

Table 6: Holding-horizon ablation and linguistic distillation shifts.